Angular contrast and rounding

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1. Quantum model, notation and sufficient two-dimensional reduction

Let `P=|p><p|` be a rank-one projector and `R=I−P` on the specified joint-probe Hilbert space. Let `ρ` be a normalized state and `0≤E≤I` a *common* binary effect, with

$$s = \mathrm{Tr}(P\rho) = \frac{131}{320}, \qquad h = \mathrm{Tr}(EP) = \frac{99}{100}, \qquad q = q(0) = \frac{81711338569043}{224995905216000}.$$

For the phase family `U_φ=P+e^{iφ}R`, write `q(φ)=Tr(E U_φ ρ U_φ†)`. Put

$$a = sh, \quad d = s(1-h), \quad t = a - q > 0, \quad b = \mathrm{Tr}(ER\rho R), \quad c = \mathrm{Tr}(EP\rho R), \quad z = 2\,\mathrm{Im}\,c.$$

The ordinary readout fixes `Re c=−(t+b)/2`. Positivity of `E` and `I−E` gives

$$|c|^2 \le \min\{ab,\ d(1-s-b)\}, \qquad 0 \le b \le 1-s.$$

These conditions are **sufficient**, not just necessary, because every admissible pair `(b,c)` is realized by the pure state `|ψ>=√s|p>+√(1−s)|u>` and the two-dimensional effect

$$E_{b,c} = \begin{pmatrix} h & \overline{c}/\sqrt{s(1-s)} \\ c/\sqrt{s(1-s)} & b/(1-s) \end{pmatrix},$$

embedded in the joint-probe space. Both determinant conditions for `E` and `I−E` are precisely the inequalities above. Their joint feasible set is convex.

Equivalently, the allowed `(b,z)` are the intersection of two disks:

$$(b-C_1)^2 + z^2 \le R_1^2, \quad (b-C_2)^2 + z^2 \le R_2^2,$$
$$C_1 = a+q,\ R_1 = 2\sqrt{aq}, \quad C_2 = -(t+2d),\ R_2 = 2\sqrt{d(1-q)}.$$

They meet at `b₀=(1−s)(1−h)` and `z=±z₀`, with

$$z_0 = \sqrt{4ab_0 - (t+b_0)^2}, \qquad \kappa = \frac{z_0}{t+b_0} = 1.775489004780283195 84\ldots.$$

2. Sharp two-sided contrast for a deterministic coherent offset

Let the intended relative phase be `π`; assume that its **only** error is a coherent offset `±δ` in the very same family `U_φ`, with the **same** `ρ`, `E` and `P` in both arms. Then, upon choosing the adverse sign of the imaginary interference,

$$\Delta(\delta) := q(\pi \pm \delta) - q(0) = A(t+b) - Bz, \qquad A = 1+\cos\delta, \quad B = \sin\delta$$

for `0≤δ≤π` (the signs are interchangeable because `z` can take either sign). Set `H=√(A²+B²)=2cos(δ/2)`.

Sharp minimum. Let

$$\delta_1 = 2\arccos[(C_1 - b_0)/R_1] = 12.7591826805379687\ldots^\circ .$$

For $0 \le \delta \le \delta_1$, the supporting point of the first disk is feasible for both, giving

$$\Delta_{\min}(\delta) = A(t + C_1) - R_1 H = 4a\cos^2(\delta/2) - 4\sqrt{aq}\cos(\delta/2).$$

For $\delta_1 \le \delta \le \delta_c$ the supporting point would leave the disk intersection; the minimum is at its shared upper point (b_0, z_0):

$$\Delta_{\min}(\delta) = A(t + b_0) - Bz_0.$$

Its zero occurs sharply at

$$\delta_c = 2\arctan(1/\kappa) = 58.77860795856966018\ldots^\circ .$$

Every compatible state/effect yields strictly positive contrast for $|\delta| < \delta_c$. At δ_c an admissible two-dimensional **projective** effect gives zero, so this is an open condition.

New sharp upper endpoint. On the same full interval $0 \le \delta \le \delta_c$, maximizing the contrast over the *second* disk gives the simultaneously attainable bound

$$\boxed{\Delta_{\max}(\delta) = A(t + C_2) + R_2 H = 4\sqrt{d(1-q)}\cos(\delta/2) - 4d\cos^2(\delta/2).}$$

Its supporting point is

$$(b, z) = \left(C_2 + R_2\frac{A}{H},\ -R_2\frac{B}{H}\right).$$

This point belongs to the first disk whenever $b \ge b_0$. For the supplied inputs it remains feasible through $\delta = 113.20968269065858...^\circ$, well beyond δ_c. The upper formula therefore holds throughout the guaranteed-positive interval. The constructed two-dimensional $E_\{b,c\}$ has $0 \le E \le I$ and attains the rate. The image of the convex disk intersection under the linear contrast is a full interval, so **every intermediate contrast is attainable** as well.

Offset	Sharp lower increase (percentage points)	Sharp upper increase (percentage points)
0°	8.653592287309	18.786135457214
10°	8.006123191762	18.720856010456
20°	6.398293220245	18.525231209691
30°	4.697644815950	18.199908655488
45°	2.168778595748	17.471285101295
55°	0.572299788190	16.827620196169
58°	0.116299082295	16.610292664719

These are percentages of theoretical binary *probabilities*, not percentages measured on devices. The decimal entries in this table are rounded descriptions, not lower/upper decimal certificates. The companion verifier emits **downward-rounded lower certificates** for Δ_min separately.

3. Sharp calibration criterion for an independently varying phase offset

Suppose one fixed pre-phase conditional ρ and one fixed E apply to all trials in this comparison, and the distribution of actual offsets ε is transferable to that same response independently of hidden state/effect changes. Define the *population* moments

$$C = \mathbb{E}\cos\varepsilon, \qquad S = \mathbb{E}\sin\varepsilon, \qquad C^2 + S^2 \le 1.$$

No values of these moments are assumed known from experiment. Averaging the fixed response gives exactly

$$\overline{\Delta} = (1 + C)(t + b) - Sz.$$

Put A=1+C, B=|S|, H=√(A²+B²) and b_* = C₁−R₁ A/H when H>0. The **sharp minimum** over all state/effect pairs compatible with the three supplied population inputs is

$$\boxed{M(C,S) = \begin{cases} A(t + C_1) - R_1 H, & b_* \le b_0, \\ A(t + b_0) - Bz_0, & b_* > b_0. \end{cases}}$$

If A=B=0 (the phase is reversed relative to the intended reflection with probability one), take M=0. The proof is the same supporting-line optimization over the two-disk intersection, and both branches are attainable.

The exact **strict sign criterion** is

$$\boxed{1 + C > \kappa|S|.}$$

Necessity as well as sufficiency follows because max z/(t+b)=κ at the disk intersection, and t+b>0 throughout the feasible set. In the symmetric case S=0 the sharp lower contrast simplifies to

$$\boxed{M(C,0) = \frac{1+C}{2}\, 4(a - \sqrt{aq}).}$$

Simultaneous, independently justified confidence bounds C≥C_L and |S|≤S_U therefore give the sufficient physical sign criterion 1+C_L>κ S_U *only when* the common-state/effect and transfer hypotheses have been independently established. The rectangular bound can be conservative if it includes moment pairs that no phase distribution can realize.

Nontransfer control. Two hidden conditional subpopulations can have the same s,h,q but opposite imaginary interference coefficients. If positive and negative angle errors are systematically correlated with these subpopulations, their pooled error histogram can have S=0 while both arms have the adverse response. At the sign boundary their average contrast can be zero. Thus an unlabelled marginal angle histogram is **not** evidence for the independent-transfer condition, and may not be inserted into M(C,S) without justification.